Reg. No: 192511020

SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student :

Programme: **B.E / B.Tech**

Course Code: **CSA15**

Course Name: **Cloud Computing and Big Data Analytics**

Duration: **1:30 Hr**

Slot : A

Faculty : Dr.Arul Raj A.M

Date:

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
Total				50	

Gr. Sanjana		
Student Signature	**Faculty Signature**	**Program Director**

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices.

(10) (CO1) (BL3)

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

Given data

(10) (CO2) (BL3)

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

CIA 1 Test

NAME: Sanjana G
Reg no: 192510020
Course: Cloud Computing
Course Code: CSA1520

1) Introduction:

Cloud computing provides computing resources and software through the Internet. Depelopments in hardware, virtualization and Internet technologies have enabled Organizations to use cloud services efficiently.

(i) Developments in Hardware and Internet Software.

* Virtualization
* Multi-core processor
* High-Speed storage
* APIs
* Automation

(ii) Cloud service Models.

Requirement	Model	Explanation
Existing application with OS control	Iaas	provides VM, storage and Network
New application without managing servers	Paas	provides development tools and runtime environments.
Ready-made CRM through browser	Saas	provides complete software managed by the Provider.

IaaS – Infrastructure as a service : provides virtual servers, storage and internet. Gives customers images the OS and applications.
Ex: Amazon EC2.

PaaS – Platform as a service : provides a platform for application development and deployment. The provider manages servers and runtime environment.
Ex: Google App Engine.

SaaS – Software as a service provides ready to use applications through the Internet. Ex: Salesforce etc.

3 XaaS Model

XaaS means Everything as a service. It provides additional IT resources as services.
Ex:
- Storage as a service
- Database as a service
- Backup as a service

Conclusion:

IaaS, PaaS and SaaS provide different levels of control and management. XaaS extends cloud computing by offering additional services, reducing costs and improving flexibility.

2) Given Data:

Old servers = 15, CPU utilization = 25%, Old server power = 350 W
New host power = 450W, Target utilization = 65%,
N+1 = one additional host, Electricity tariff = ₹9/kWh
Annual operating hours = 8,760.

(a) Number of Hosts

$15 \times 25\% = 3.75$

$N = \frac{3.75}{0.65} = 5.77$

6 active hosts

N+1 ⇒ 6 + 1 ⇒ 7 total hosts

(b) Consolidation Ratio

$\frac{15}{6} = 2.5$

2.5 : 1

(c) CPU utilization

$\frac{3.75}{7} \times 100 = 53.57\%$

After one host fails.

$\frac{3.75}{6} \times 100 = 62.5\%$

(d) Annual Electricity Cost Saving

$15 \times 350 = 5250\ W = 5.25\ KW$

Cold = $5.25 \times 8760 \times 9$

= ₹ 4, 13, 910

New system:

7 × 450 = 3150 W = 3.15 kW

$C_{new} = 3.15 \times 8760 \times 9$

₹ 2,48,346

Annual saving:

₹ 4,13,910 − ₹ 2,48,346

₹ 1,65,564

Conclusion:

Consolidation reduces 15 physical servers to 7 hosts and provides N+1 fault tolerance. The annual electricity cost saving is ₹ 1,65,564.

Assumption:

All hosts are powered on and each new host provides equal CPU capacity to the old servers.

3) (a) CPU and Memory over-commitment Ratios

Total allocated vCPUs:

12 × 3 = 36

CPU ratio:

36/24 = 1.5:1

Available memory:

96 − 8 = 88 GB

(b) Memory sufficiency

Normal operation:

12 × 5 = 60 GB

free memory:

$88 - 60 = \boxed{28\ GB}$

Memory is sufficient.

Peak operation:

$(8 \times 8) + (1 \times 6) = 88\ GB$

free memory = 0 GB

Memory is Exactly sufficient, but no free memory remains.

(C) CPU Sufficiency

normal operation:

$12 \times 3 \times 35\% = 12.6$

$\frac{12.6}{24} \times 100 = \boxed{52.5\%}$

CPU is Sufficient.

Peak operation:

$12 \times 3 \times 75\% = 27$

$\frac{27}{24} \times 100 = \boxed{112.5\%}$

CPU demand exceeds capacity by 3 cores.

∴ CPU is insufficient during peak operation.

d) performance and security Issues with Remedies.

Issues	Remedies
CPU contention memory shortage Data leakage	Add CPU cores Increases RAM use encryption and network isolation.

Conclusion:

The host is sufficient during normal operation. At peak, CPU demand exceeds capacity and memory is fully utilized. Increasing resources and maintaining security controls can improve performance.

4) Introduction:

Cloud federation allows two independent universities to share computing resources and services.

(i) Conventional Cloud Architecture

- Consists of servers, storage, networking, hypervisors
- Supports resource sharing and VM Management

(ii) SDDC

SDDC Manages Computing, storage, and networking through software.

- Dynamic resource allocation.
- Automated provisioning.
- Workload migration.

(iii) Role of SOA

SOA Enables services sharing through standard APIs.

- University A provides research services.
- University B provides digital - Library Services.

Challenges:

* Security and interoperability.
* Resource Allocation
* Identity management.
* SLM management.

Privacy Risks

* Presence information may reveal User activity.
* Federated Identity may cause Identity theft or unauthorized access.

Safeguards:

- Use Encryption and multi-factor Authentication.
- Implement role-based access control.
- Establish data-sharing agreements.

Conclusion:

Cloud federation enables resource sharing. SDIC manages resources / SOA enables service Communication and security safeguards protect user privacy.

5) (a) Daily Data Volume and Bandwidth

Clips per Camera per day:

$$\frac{24 \times 60}{3} = 480$$

Total clips:

$$150 \times 480 = 72,000$$

Daily data volume:

72,000 × 4 = 288,000 MB

[288 GB / day]

Average bandwidth:

$\frac{288 \times 8}{86400}$ = [26.67 Mbps]

(b) Monthly storage cost

Monthly Storage:

288 × 30 = 8640 GB

Cost:

8640 × 1.50 = [₹12,960]

(c) After edge filtering.

Only 25% of clips are uploaded.

Daily data:

288 × 0.25 = [72 GB]

Average bandwidth:

26.67 × 0.25 = [6.67 Mbps]

Monthly storage cost:

72 × 30 × 1.50 = [₹ 3,240]

Monthly saving:

12,960 − 3,240 = [₹9,720]

(d) Lambda Invocation Cost

Monthly invocation:

72,000 × 30 × 0.25 = 540,000

Cost:

540,000 × 0.00015 = ₹81

Rest API and Object Storage:

Rest API: Enables video clips to be Uploaded to cloud storage using HTTP requests.

Object storage:

Suitable for large video files because. It provides scalable, durable, and cost-effective storage.

Conclusion:

Edge filtering reduces data volume from 288 GB to 72 GB day and saves ₹9,720 monthly in storage costs.